FRP 2026 AI ENGINEERING TRACK DAY 7

Share what you found

A paper, a page and a talk

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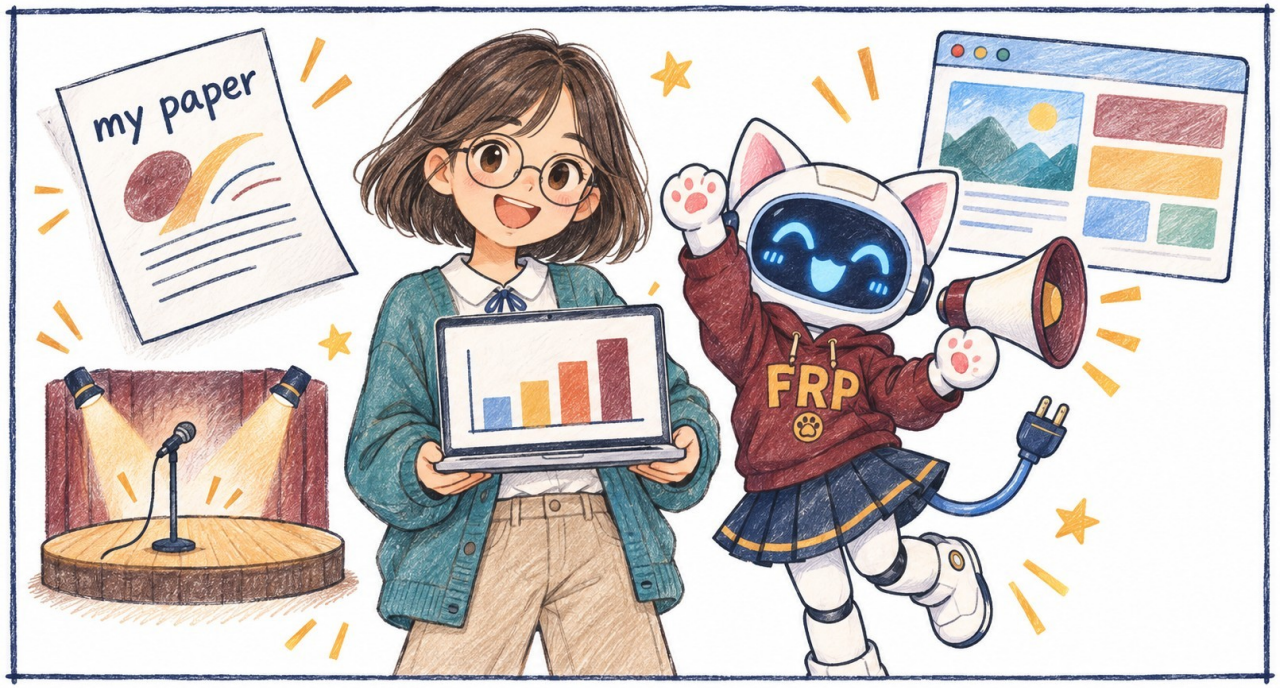
ISSAI, Nazarbayev University, 30 October 2026

You found something new.
You tell nobody.

Did it happen?

Technology without users is useless.

Research is done when it is shared



Today: a paper, a page and a talk. Honest ones.

START

FROM DAY 6

Today's example: our own discovery

Clean text: Kazakh Wikipedia

Ұзақ жылдар педагогтік қызмет атқарған.

“Worked as a teacher for many years.”

Web junk, mixed into half

Біз cookie файлдарын қолданамыз. Келісемін ✓

ЖЕҢІЛДІК!!! ЖЕҢІЛДІК!!! Қазір сатып ал!!! 50% ЖЕҢІЛДІК!!!

“We use cookies. I agree.” “SALE!!!”

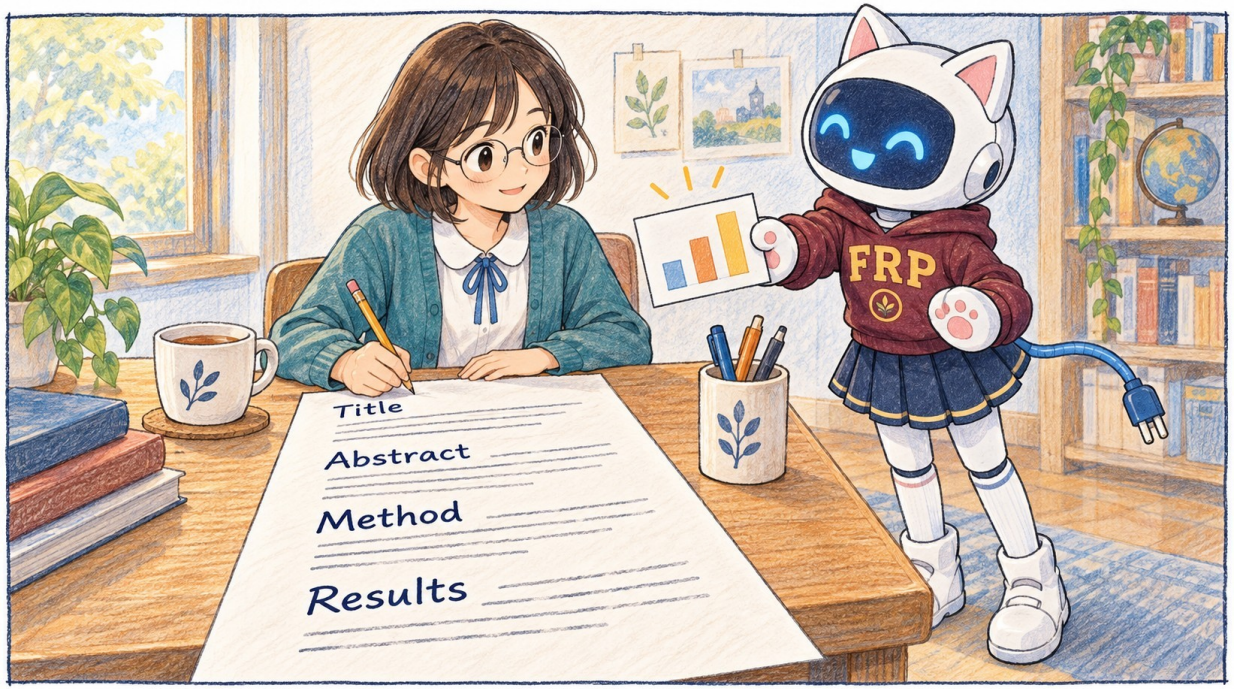
Same tiny model. After clean text: junk in 0 of 50 texts. After dirty text: 37 of 50.

PART A: THE PAPER

The paper

A story with proof.

A paper is a story with proof



The story: question, work, finding. The proof: numbers anybody can check.

The parts of a paper: one question each

Title	what is it about?
Abstract	the whole story in 4 sentences
Introduction	why does it matter?
Method	what exactly did you do?
Results	what did you find, with numbers?
Limitations	where can it be wrong?
Conclusion	what should people remember?
References	whose work did you use?

NEW WORD

abstract = the whole paper in 4 sentences

DAY 6 EXPERIMENT, 2 PAGES, LATEX

NEW WORD

baseline = what you compare your result with

Our example paper

title

what is new

abstract

4 sentences

results

with a baseline

Junk In, Junk Out: How Web Junk Changes a Tiny Kazakh Language Model

Abstract

Language models learn from text collected on the web, and web text contains junk: menus, cookie banners and spam. We trained the same tiny language model (10.9 million parameters) twice on Kazakh Wikipedia: once on clean text, and once with half of the text replaced by web junk. The model trained on dirty text wrote junk in 37 of 50 texts, and the clean model in 0 of 50; its loss on clean text was also worse (0.94 vs. 0.88). Cleaning the data is therefore not a detail: the model copies what it reads.

Training text	Loss on clean text ↓	Texts with junk (of 50) ↓
Clean (baseline)	**0.88**	**0**
Dirty (50% junk)	0.94	37

Copy it: paper_example.tex is in your workspace.

ACTIVITY: HANDS UP, A OR B

Which title is better?

A My AI project

B Junk In, Junk Out: How Web Junk Changes a Tiny Kazakh Language Model

A Can a Small Model Learn to Count Letters from a Score Alone?

B Experiments with models

A A study of tokenizers

B 16,000 New Kazakh Tokens Cut Kazakh Text by 43 %

THE ANSWERS: B, A, B

Good titles say what is new

✓ **Junk In, Junk Out: How Web Junk Changes a Tiny Kazakh Language Model**

✓ **Can a Small Model Learn to Count Letters from a Score Alone?**

✓ **16,000 New Kazakh Tokens Cut Kazakh Text by 43 %**

> **A question, or the result with a number. Never only the topic.**

An abstract = 4 sentences

1 problem	Language models learn from web text, and web text contains junk.
2 what we did	We trained the same tiny model twice on Kazakh Wikipedia: on clean text, and with half of it replaced by web junk.
3 the number	The dirty model wrote junk in 37 of 50 texts, the clean model in 0 of 50.
4 why it matters	Look at your data: the model copies what it reads.

ACTIVITY IN PAIRS, 6 MINUTES

Write your 4 sentences

1 problem

Who has which problem?

2 what you did

What did you build or test?

3 the number

What did you measure?

Compared with what?

4 why it matters

Who is it good for?

No number yet? Right answers of 20 tests, seconds to answer, 4 of 5 friends finished.

SWAP WITH ANOTHER PAIR, 2 MINUTES

Read it like a reviewer

1. **Is there a number?**

2. **Is it clear what is new?**

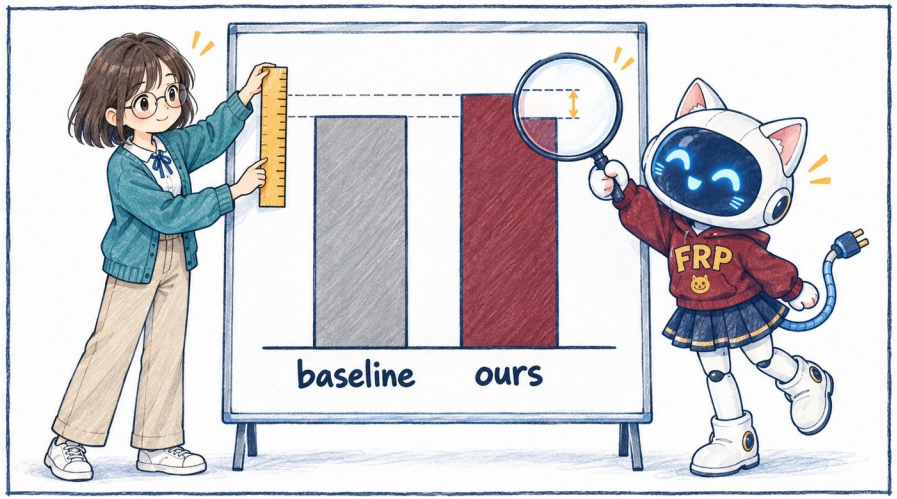
3. **Would your grandmother understand sentence 1?**

> **This is what reviewers do with real papers.**

Results people trust

A number needs a baseline.

Always compare with a baseline



Our baselines this week: Qwen before practice, the original Qwen, the clean model.

Better than what? Always say it.

An honest table

Training text	Loss on clean text ↓	Texts with junk ↓
Clean (baseline)	**0.88**	**0 of 50**
Dirty (50 % junk)	0.94	37 of 50

Same model, same steps; only the training text differs.

> **Same test, best in bold, an arrow, a caption that says what changed.**

One chart, one message

Texts with junk, of 50 (lower is better)



Train on junk, and the model writes junk: 37 of 50 texts.

Report the bad news too

Our Kazakh model

hardest science test:

84.3 → 76.3

we wrote it down

Our class teams

second run: 50, 51, 50 %

numbers move

run more than once

Every paper

has a Limitations section

where can it be wrong?

Hiding bad news is how people stop trusting you.

Spot the problems

“Our app is much better: 95 %!”

What is missing? Hands up.

Never write a sentence like this in your paper.

A sentence you can trust

"The same tiny model wrote junk in **37 of 50** texts after training on 50 % junk text, and in **0 of 50** after clean text."

Better than what? the clean model

Of how many? 50 texts each

One try or many? one run each: in Limitations

Who checked? a script counted the junk

A baseline, a count, the number of tries, a checker.

PART C: WRITE IT IN OVERLEAF

Write it in Overleaf

LaTeX in your browser, with AI as a helper.

CODE

LaTeX: plain text in, a beautiful PDF out

1 You write: plain text with commands

day7 > paper > latex_side.tex

```latex
\documentclass{article}
\title{Junk In, Junk Out}
\author{Aru, FRP 2026}
\begin{document}
\maketitle
\begin{abstract}
Web text contains junk: menus and spam.
\end{abstract}
\section{Method}
We trained the same tiny model twice.
\section{Results}
Junk in 37 of 50 texts; clean model: 0.
\end{document}
```

2 You get: the PDF

Junk In, Junk Out

Aru, FRP 2026

October 15, 2026

Abstract

Web text contains junk: menus and spam.

1 Method

We trained the same tiny model twice.

2 Results

Junk in 37 of 50 texts; clean model: 0.

Why LaTeX? One clean look, easy formulas, journals use it.

WATCH: I PASTE AND COMPILE ON THE BIG SCREEN

Overleaf + AI, the right way

AI can help you

"make my abstract 4 sentences"

"check my English"

"make a LaTeX table from these numbers"

AI must never

invent your results

invent your references

(Day 2: it invents, and sounds sure)

Now: watch me paste and Recompile. Your account and your paper: after lunch, with a mentor.

Break, 10 minutes

Next: your own project page

PART D: THE PROJECT PAGE

The project page

One minute, on a phone.

What a project page has



Title, names, 3 buttons (paper, code, demo), a picture, abstract, results, cite.

ONE FILE: INDEX.HTML, IN YOUR WORKSPACE

Our example page



Live: kitan-lab-pixel.github.io/frp-example-page

CODE

Ask the agent for your page

1 Type it in OpenCode: the 8 parts of the page

```
/plan Make a project page for my project in one file, index.html.
Parts: a big title, my name, three buttons (Paper, Code, Demo), one
big picture, my abstract (4 sentences), a results table with a
baseline, one chart, and a "Cite" box. White background, big text,
and it must work on a phone.

Build · Qwen3.8-27B FRP Qwen
```

2 The agent's plan, in 1 minute, with your real numbers

```
Plan

1. Simplest version that works: page/index.html with title "Junk In, Junk Out", your name, the 4-sentence
   abstract, and the results table with your real numbers. White background, big text.
2. Buttons + picture: add the 3 buttons (Paper → day7/paper, Code → day6 scripts, Demo → Kazakh Speak page) and
   the big picture (teaser.png copied next to index.html).
3. Chart: one chart (bars: loss and junk texts for clean vs dirty) built from the table, no libraries.
4. Cite + phone check: the Cite box, then open the page in a narrow window and fix anything that breaks on a
   phone.

If this is OK, type /build.

▣ Build · Qwen3.8-27B · 59.4s
```

Then /build, open it, and check it on your phone.

Like every result.

Publish it: GitHub Pages

“Publish my page folder as a NEW public repository <project>-page on GitHub Pages. Keep my frp-2026 repository private.”

What the agent runs for you (in the terminal, logged in to GitHub):

```
cd ~/work && echo "my-page/" >> .gitignore        # keep the page out of your private repo
gh repo create my-page --public --clone && cd my-page   # a new, empty, public repo
(copy index.html + images/ into it)
git add -A && git commit -m page && git branch -M main && git push -u origin main
gh api -X POST repos/YOUR-NAME/my-page/pages -f "source[branch]=main" -f "source[path]=/"
```

A new public repo only for the page. Your work stays private.
No phone numbers, no private photos.

WORKSHOP, 12 MINUTES

Your page, online

- ☐ /plan with the parts you want
- ☐ /build, then open it in your browser
- ☐ ask the agent to publish it
- ☐ open it on your neighbour's phone

Stuck?

No GitHub yet? Ask the agent: “show my page in the browser”.

A mentor helps you publish after lunch.

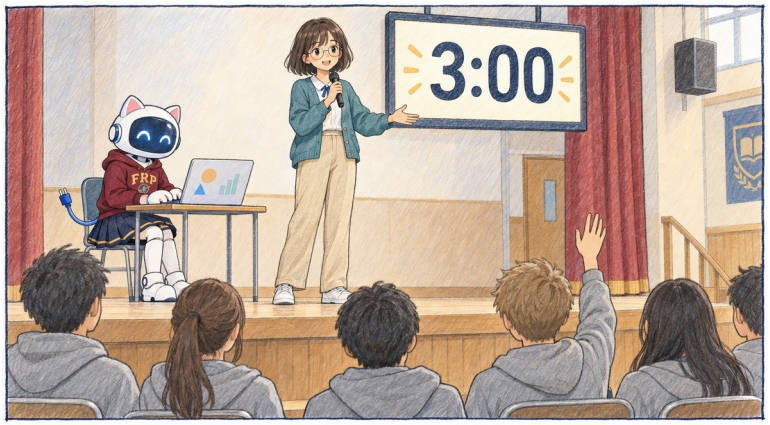
Done = your page opens in your browser. With GitHub: also on a phone.

PART E: THE TALK

The talk

3 minutes and a live demo.

A 3-minute talk



hook	problem	what I built	demo	my number	next
20 s	30 s	40 s	50 s	20 s	20 s

Practise with a timer: 3 minutes is shorter than you think.

Live demo rules

Before

rehearse 3 times

a backup video

one tested input

During

big font

say what to look at

slow

If it fails

smile

“this is why we have a backup”

play the video

Everybody remembers a calm presenter.

Slides for a talk

✗ a wall of text

Results and discussion of our experiments

- We trained a character-level transformer (6 layers, width 384, 10.9M parameters) on Kazakh Wikipedia.
- The clean run used 60 million letters. In the dirty run every 2,000-letter chunk was replaced by web junk with probability 0.5 (menus, HTML, cookie banners, spam, broken encoding).
- Training ran 12,000 steps (197 million letters seen) on one NVIDIA V100 GPU, about 7 minutes per run.
- Loss on 300 hidden clean articles (lower is better): clean model 0.88, dirty model 0.94.
- We let each model write 50 texts and counted the texts that contain junk.
- Clean model: 0 of 50 texts with junk. Dirty model: 37 of 50 texts with junk.
- The dirty model also learned the clean text less well, because the junk took half of its practice.
- Limitations: we trained each model once; a second run can change the numbers a little.
- Our junk is simpler than real web junk, and the model is very small, so large models may differ.
- Conclusion: look at your data before you train on it, because the model copies what it reads.

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✓ one idea, big, a number

PART B: RESULTS

One chart, one message

Texts with junk, of 50 (lower is better)



Train on junk, and the model writes junk: 37 of 50 texts.

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If people read, they do not listen.

ACTIVITY IN PAIRS, 8 MINUTES

Your 30-second pitch

"I made … for …. It …. My number is …."

My number, for example: 17 of 20 test questions right, 2 seconds to answer, 4 of 5 friends finished the task.

Partner: holds the timer, says 1 thing that was great and 1 thing to change.

> **Then swap. 2 or 3 of you on stage.**

PART F: BE FAIR

Be fair

Credit, consent, honesty.

NEW WORD

consent = a person's yes before you use their voice or face

THE LICENCES OF OUR DATA THIS WEEK

Credit what you use

Kazakh Wikipedia	CC BY-SA 4.0: name it, share alike
Abai's poems	public domain
SmolTalk, Qwen3	Apache 2.0: free, with credit
KazMMLU	CC BY-NC 4.0: not for selling

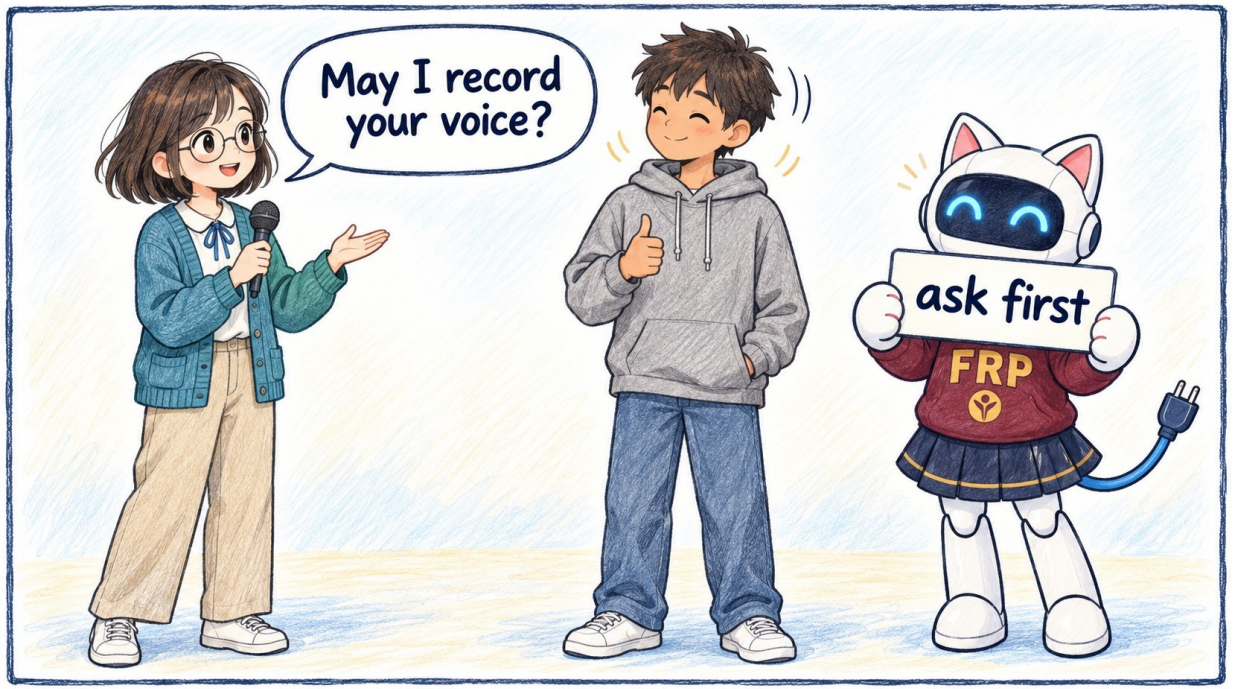
NEW WORD

licence = the rules for using someone's work

Example line: Kazakh Wikipedia (CC BY-SA 4.0): name it, share alike.

Your task: in your references, write each dataset and model you used, and its rule.

Voices and faces: ask first



Ask first, and say where it will be public. No private data on a public page.

Honest work

Never	Keep	Say
make up a result	your failures in the paper	where AI helped you

“We used an AI agent to write the code and to check the English.” Normal, and honest.

Tonight and Demo Day

After lunch

paper in Overleaf (PDF)

page online

talk practised twice

Tonight

backup video of your demo

Demo Day

3 minutes

a demo, a number

your page on the screen

I am proud of what you built this week.

3 THINGS TO REMEMBER

1. Share it, or it did not happen: a paper, a page, a talk.
2. A number needs a baseline. Bad news belongs in the paper too.
3. Give credit, and ask first.

Sources

Example research: our Day 6 runs on the ISSAI FRP server, 14 Oct 2026 (day6-training/2_facts.md A1, A2, A5, B9).

Example paper: 4_visuals/demos/paper/paper_example.tex (compiled with tectonic). Example page: demos/project_page/index.html.

Licences: Kazakh Wikipedia CC BY-SA 4.0; Abai, Wikisource (public domain); SmolTalk Apache 2.0; Qwen3 Apache 2.0; KazMMLU CC BY-NC 4.0 (MBZUAI).

Overleaf: overleaf.com. GitHub Pages: docs.github.com/pages (REST: POST /repos/{owner}/{repo}/pages).